

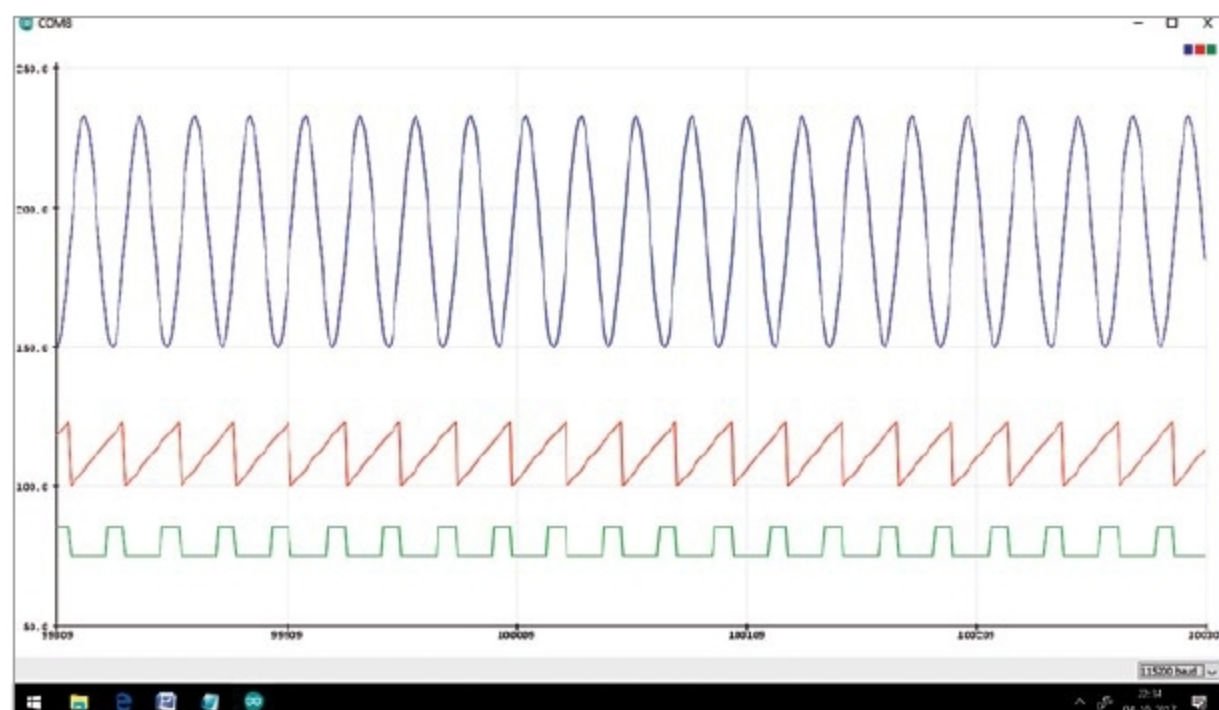


Fig. 3: Signal waveforms on serial plotter

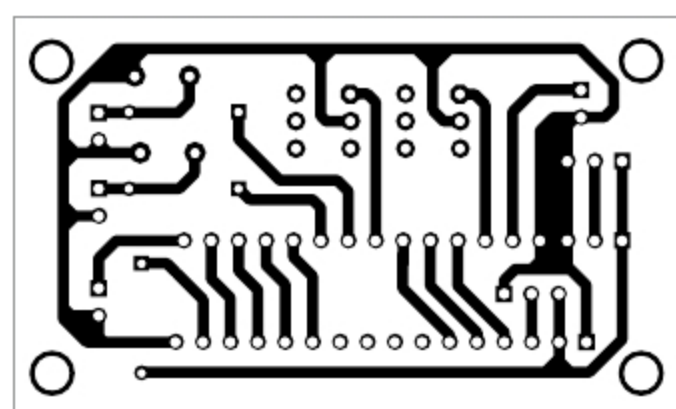


Fig. 4: Actual-size PCB layout for the frequency generator

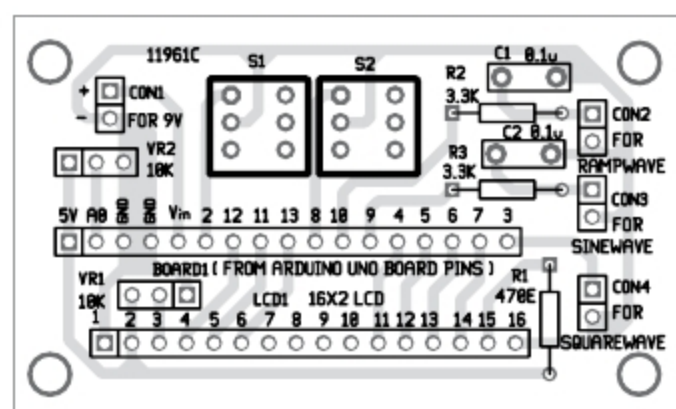


Fig. 5: Components layout for the PCB

shown in Fig. 2.

These waveforms are synthesized using Timer 0 and Compare-Match interrupt control functions of Arduino microcontroller (ATmega328). Timer 1 of ATmega328 is programmed at 10kHz to generate PWM output signals. The program (freq_gen.ino) contains registers of the ATmega328 microcontroller

configured as per datasheet. The software (freq_gen.ino) is written in Arduino programming language and compiled using Arduino IDE software.

Switch S2 connected to pin 8 of Board1 is used for changing the frequency range. Two frequency ranges are designed in the program: 30 to 250Hz and 250 to 2500Hz to cover the medium audio frequency range. These signals from CON2 through CON4 can be viewed on an oscilloscope.

The program is to be uploaded into the Arduino board from the PC using Arduino IDE software. With USB cable connected between Arduino and PC, the frequency value can be viewed on LCD1. Switch S1 connected to pin 2 of Board1 is used for changing the display between LCD1 and serial plotter on Arduino IDE.

If pin 2 is grounded, the waveforms (sine, square and ramp) can be viewed on the serial plotter like a digital oscilloscope (refer Fig. 3).

Each waveform has a magnitude of about 5V. So the sine wave changes from 0 to 5V and does not go to negative.

LCD1 connections are made on the



PCB using 16-pin Berg strip. A 470-ohm resistor (R1) connected to pin 15 is used to light LCD1. 5V for the LCD is taken from Arduino board's pins 5V and Gnd.

A USB cable is used to connect Arduino to a PC or laptop. After uploading the program (freq_gen.ino), the Arduino board and LCD1 can be powered by a 9V adaptor/battery.

Construction and testing

An actual-size PCB layout for the frequency generator is shown in Fig. 4 and its components layout in Fig. 5. After assembling the circuit on the PCB, upload the source code (freq_gen.ino) to Arduino board. Disconnect Board1 from computer and connect it to 9V supply across CON1. You can view the generated frequency value on LCD1 by opening switch S1, or check different signal waveforms on the serial plotter by closing S1. **EFY**



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